

Module -1

DC Circuit Analysis and Theorems

Dr. Sathya P
Associate Professor
SENSE

Nodal Analysis

(Based on KCL)

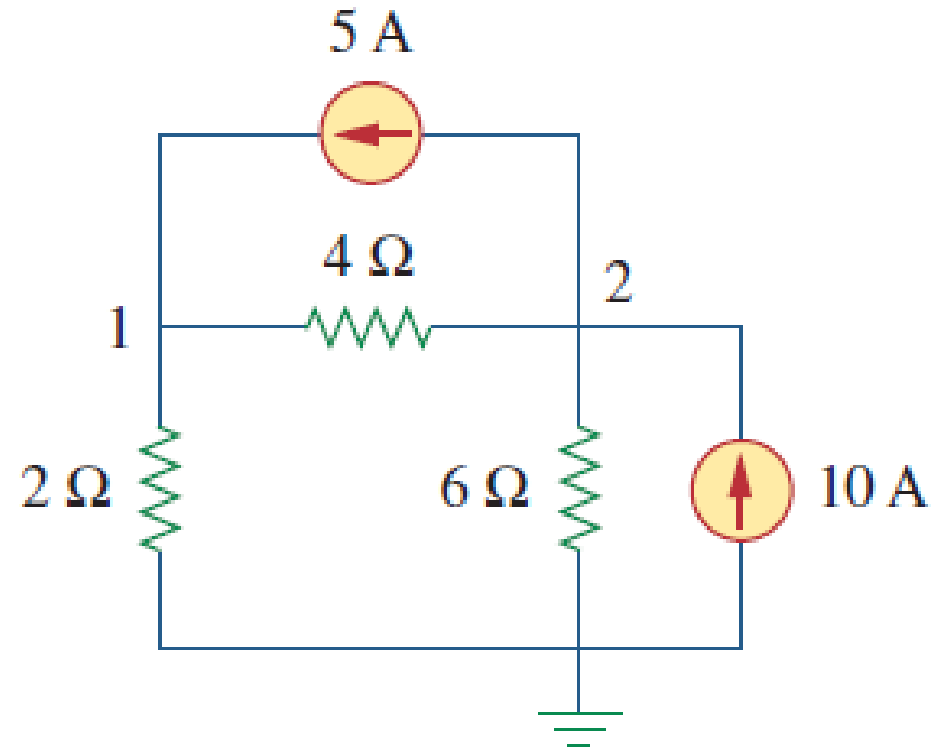
Nodal analysis method is used to find the node voltages in a circuit which are considered as the circuit variables.

Steps:

1. Select a node as the reference node. Assign voltages $V_1, V_2, \dots, V_{n-1}$ to the remaining $n-1$ nodes.
2. Apply KCL to each of the $n-1$ non-reference nodes. Use ohm's law to express the branch currents in terms of node voltages.
3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

Tutorial - 1

Calculate the node voltages in the circuit.



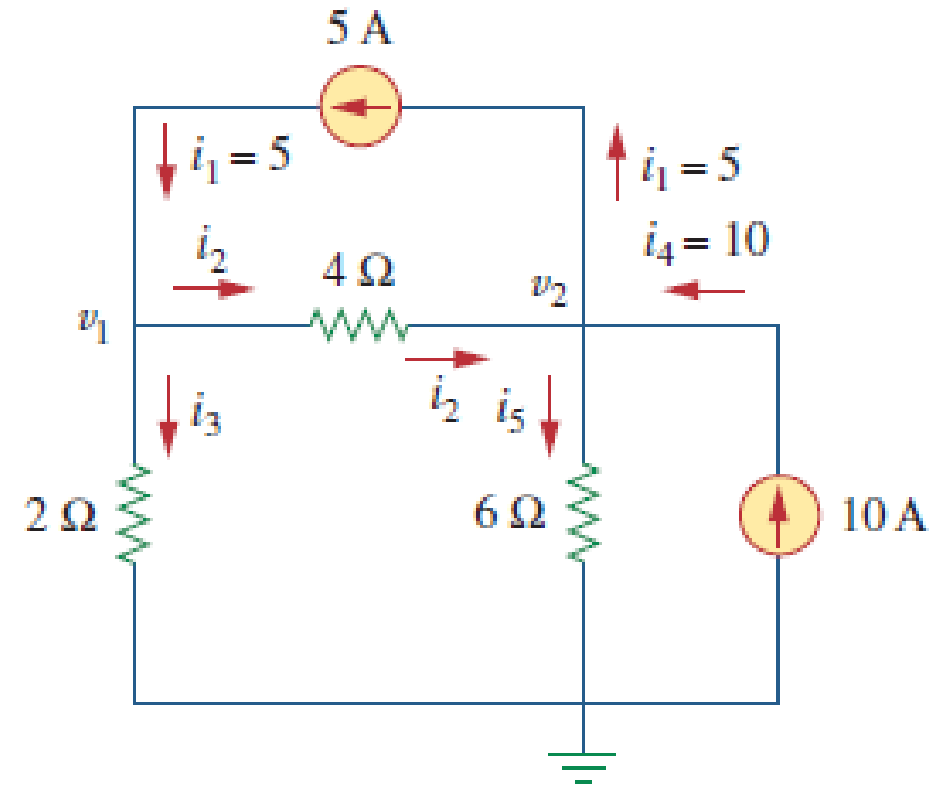
- Solution:

At node 1, applying KCL and Ohm's law gives

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

$$20 = v_1 - v_2 + 2v_1$$

$$3v_1 - v_2 = 20$$



At node 2, we do the same thing and get

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

$$3v_1 - 3v_2 + 120 = 60 + 2v_2$$

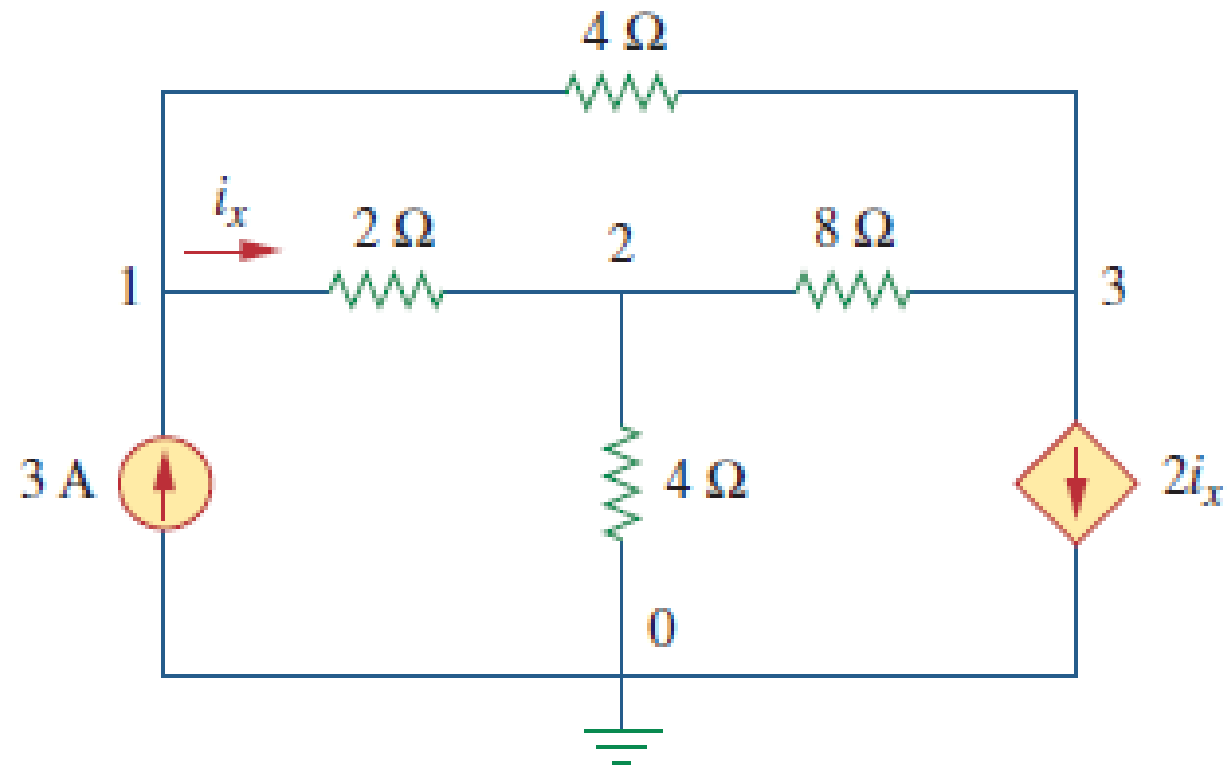
$$-3v_1 + 5v_2 = 60$$

$$4v_2 = 80 \quad \Rightarrow \quad v_2 = 20 \text{ V}$$

$$3v_1 - 20 = 20 \quad \Rightarrow \quad v_1 = \frac{40}{3} = 13.333 \text{ V}$$

Tutorial - 2

Determine the node voltages in the circuit.



- Solution:

At node 1,

$$3 = i_1 + i_x \quad \Rightarrow \quad 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$$

Multiplying by 4 and rearranging terms, we get

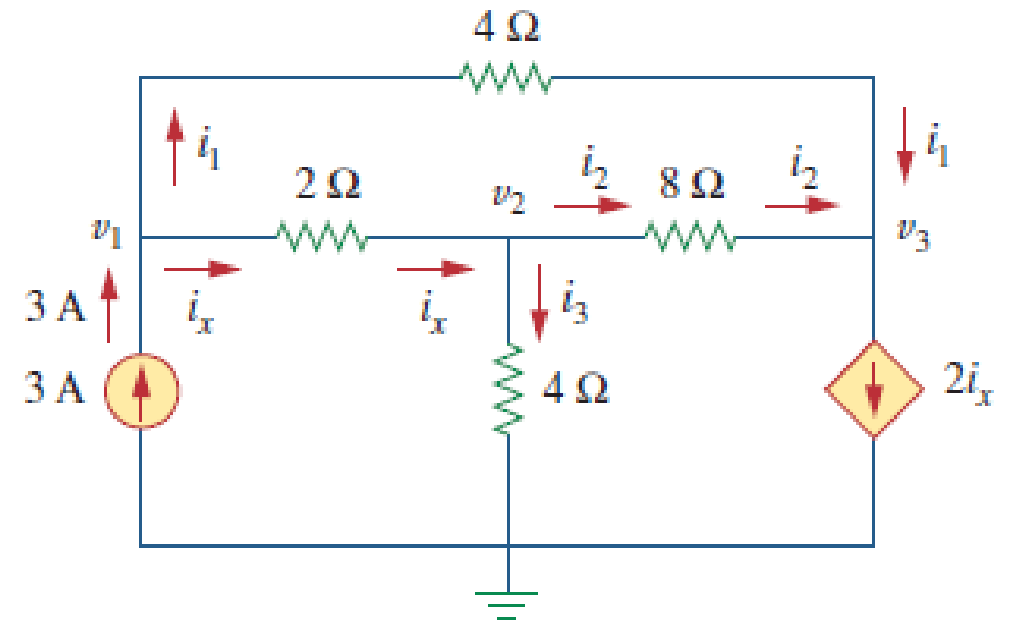
$$3v_1 - 2v_2 - v_3 = 12$$

At node 2,

$$i_x = i_2 + i_3 \quad \Rightarrow \quad \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$

Multiplying by 8 and rearranging terms, we get

$$-4v_1 + 7v_2 - v_3 = 0$$



At node 3,

$$i_1 + i_2 = 2i_x \quad \Rightarrow \quad \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$

Multiplying by 8, rearranging terms, and dividing by 3, we get

$$2v_1 - 3v_2 + v_3 = 0$$

$$5v_1 - 5v_2 = 12$$

$$v_1 - v_2 = \frac{12}{5} = 2.4$$

$$-2v_1 + 4v_2 = 0 \quad \Rightarrow \quad v_1 = 2v_2$$

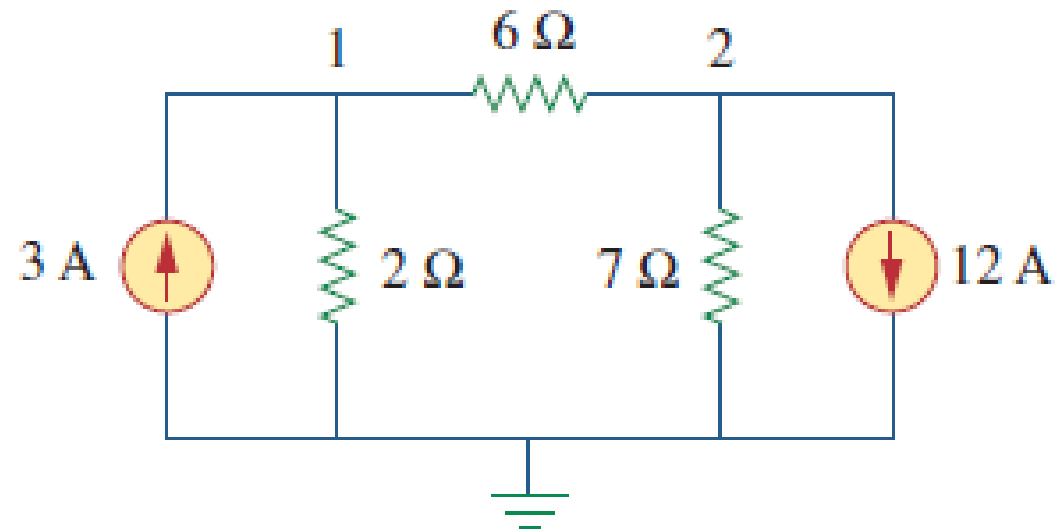
$$2v_2 - v_2 = 2.4 \quad \Rightarrow \quad v_2 = 2.4, \quad v_1 = 2v_2 = 4.8 \text{ V}$$

$$v_3 = 3v_2 - 2v_1 = 3v_2 - 4v_2 = -v_2 = -2.4 \text{ V}$$

$$v_1 = 4.8 \text{ V}, \quad v_2 = 2.4 \text{ V}, \quad v_3 = -2.4 \text{ V}$$

Tutorial - 3

Obtain the node voltages in the circuit



Answer: $v_1 = -6\text{ V}$, $v_2 = -42\text{ V}$.